# STM32F401RE Evaluation Board

## Progress Evaluation

*Mixed Traces PCB Design Competition*

This document summarizes the team's current progress toward a compact STM32F401RE / STM32F401RET6 evaluation/development board. The design is being developed in KiCad as a two-layer PCB within the 100 mm × 100 mm maximum board size.

## 1. Selected MCU & Design Concept

Selected MCU: STM32F401RET6 in the LQFP64 package.

Design concept: a compact general-purpose evaluation/development board with regulated 3.3 V power, SWD programming/debugging, reset and user controls, USB, UART, SPI, I²C, and GPIO expansion.

The STM32 NUCLEO-F401RE is used as the primary functional reference; the team's schematic, component placement, and PCB routing are independently designed.

## 2. Block Diagram

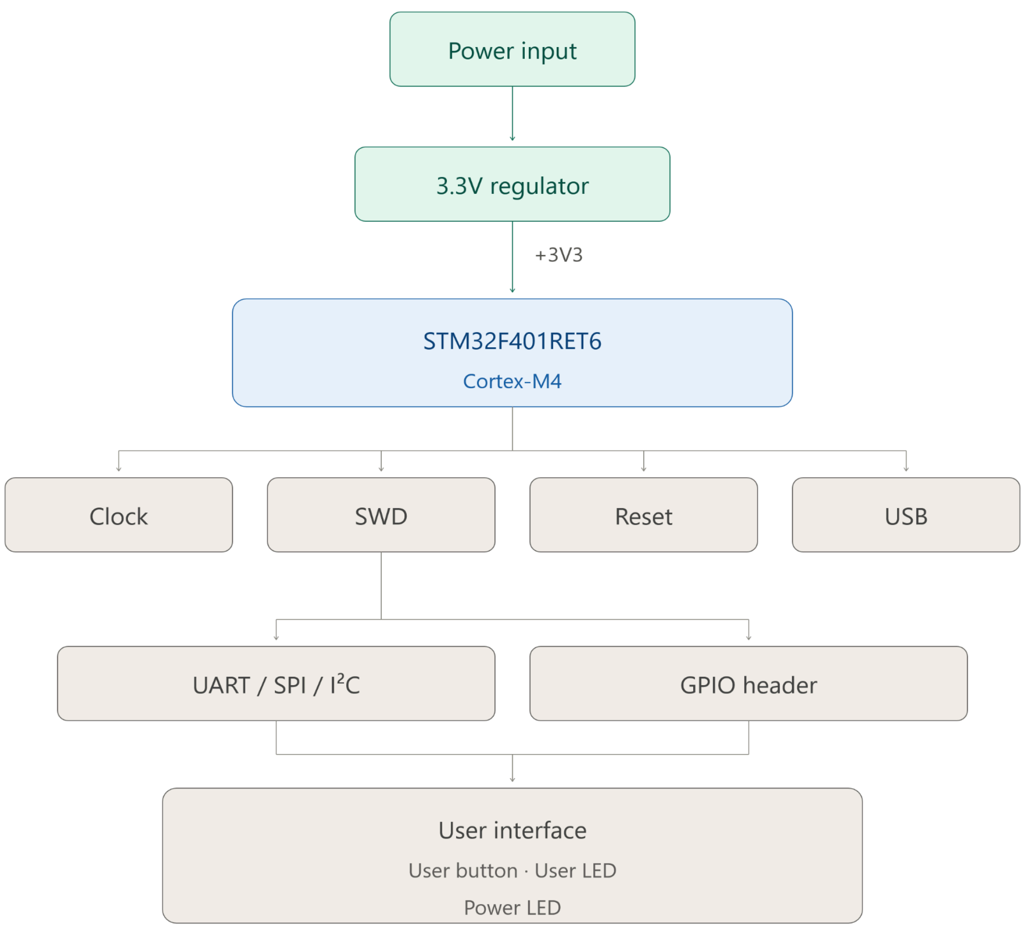

*Figure 1. Proposed STM32F401RE evaluation-board block diagram*

## 3. Schematic Progress

| Section | Status | Current progress |
|---|---|---|
| STM32F401RET6 LQFP64 | Completed | MCU symbol and LQFP64 footprint selected. |
| VDD / VSS | Completed | Four VDD pins connected to +3V3; VSS connected to GND. |
| VDD decoupling | Completed | Four 100 nF capacitors added. |
| VDD bulk filtering | Completed | 4.7 µF bulk capacitor added. |
| VDDA / VSSA / VREF | Completed | 1 µF and 10 nF filtering network added. |
| VCAP1 | Completed | 4.7 µF capacitor connected from VCAP1 to GND. |
| VBAT | Completed | VBAT connected to +3V3 for the basic configuration. |
| Clock | In progress | HSE/clock implementation being evaluated from STM32 documentation. |
| Reset / BOOT0 / SWD | Planned | To be added to the MCU/programming section. |
| USB / UART / SPI / I²C / GPIO | Planned | Assigned to the communication/GPIO section. |
| Power input / regulator / UI | Planned | Assigned to the power/user-interface section. |

## 4. Component Selection Progress

- STM32F401RET6 — LQFP64 package
- C1–C4 — 100 nF MCU VDD decoupling capacitors
- C5 — 4.7 µF bulk supply capacitor
- C6 — 1 µF VDDA/VSSA capacitor
- C7 — 10 nF VDDA/VSSA capacitor
- C8 — 4.7 µF VCAP1 capacitor

Remaining selection: regulator and power-input parts, clock components, reset/boot parts, SWD connector, USB/UART/SPI/I²C connectors or interface components, GPIO headers, user button/LED, and power LED.

PCB placement and routing have not yet been started. The team is currently completing the schematic blocks and component selection before moving to PCB layout.

- Exactly 2 copper layers.
- Maximum board size: 100 mm × 100 mm.
- Planned placement priority: MCU → power/regulator → decoupling → connectors/interfaces.
- Planned final checks: routing review, ground-plane review, DRC, and usability/silkscreen review.

## 6. Major Design Decisions

- STM32F401RET6 with the LQFP64 package is the selected MCU/package.
- Main digital supply rail: +3V3.
- Four 100 nF capacitors are used for the four VDD/VSS supply pairs, with a 4.7 μF bulk capacitor.
- VDDA/VSSA reference/supply filtering uses 1 μF and 10 nF capacitors.
- VCAP1 uses a dedicated capacitor to GND.
- VBAT is connected to +3V3 because no separate backup battery/supercapacitor is currently planned.
- SWD is provided without the ST-Link portion.
- The NUCLEO-F401RE is a reference only; its PCB layout is not being directly copied.

| Member | Responsibility |
|---|---|
| Neeraj S | STM32 core, clock, reset, BOOT0, SWD, and overall PCB integration. |
| Vaibhav | USB, UART, SPI, I²C, GPIO expansion, interface pin/alternate-function research, and footprints for this section. |
| Nathan Zacharia Saby | Power input, 3.3 V regulator, power filtering/distribution, power LED, user button, user LED, relevant headers/connectors, footprints, BOM and estimated cost. |

## 7. Progress Evaluation Checklist

| Required item | Status | Evidence / next action |
|---|---|---|
| Selected MCU and design concept | ✓ | STM32F401RET6 LQFP64 and board concept defined. |
| Block diagram | ✓ | Proposed block diagram included in this document. |
| Schematic progress | ✓ | MCU power section completed; remaining blocks are in progress/planned. |
| Component selection | In progress | MCU power components selected; remaining sections to be finalized. |
| PCB placement/routing progress | Not started | PCB work follows schematic completion. |
| Major design decisions | ✓ / In progress | Package, supply architecture, interfaces and work allocation established. |